Batuhan Çelik

Applying for master's and PhD admission, Fall 2027 · Symbolic representation learning and active exploration for robots

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RESEARCH INTERESTS

Symbolic representations: how to extract them from raw sensory data in a self-supervised manner, and which inductive biases let a model learn disentangled symbols and compose them to support counterfactual reasoning. Supported by active learning to allow robots to build their own curriculum of increasing complexity.

EDUCATION

Boğaziçi University, Istanbul — B.S. in Computer Engineering — Sept 2018 – Jan 2025

Five-year programme: a one-year English preparatory year (2018–19), then the four-year B.S. · GPA 3.68 / 4.00

- **Coursework:** Deep Learning in Robotics, Machine Learning, Parallel Algorithms, Advanced Theoretical Computer Science, Reconfigurable Computing, Quantum Algorithms.
- **University entrance:** ranked 517th nationally out of approximately 2.3 million candidates, science and quantitative track, Turkish university entrance examination (YKS), 2018.
- Stayed past the standard five-year completion by choice, on the advice of Prof. Suzan Üsküdarlı, in order to keep working in the CoLoRs and SISReC labs rather than finish sooner.

PUBLICATIONS

- A. Ahmetoğlu, **B. Çelik**, E. Öztop, and E. Uğur. "Discovering Predictive Relational Object Symbols with Symbolic Attentive Layers." *IEEE Robotics and Automation Letters*, 9(2):1977–1984, Feb. 2024. doi:10.1109/LRA.2024.3350994 · publication page
- **B. Çelik**, A. Ahmetoğlu, E. Uğur, and E. Öztop. "Developmental Scaffolding with Large Language Models." *IEEE Int. Conf. on Development and Learning* (*ICDL*), Nov. 2023, pp. 396–402. First and corresponding author. doi:10.1109/ICDL55364.2023.10364374 · publication page

Published as **Batuhan Celik**; full legal name **Mehmet Batuhan Çelik**.

RESEARCH EXPERIENCE

Undergraduate Researcher — CoLoRs Lab, Boğaziçi University — Sept 2023 – Feb 2024

Funded on TÜBİTAK ARDEB 1001 project 120E274 · Day-to-day supervision from Alper Ahmetoğlu; thesis assessed by Assoc. Prof. Emre Uğur

Intrinsic Motivation for Deep Symbol Learning (*B.S. thesis*) — *project page*

- Replaced uniform random exploration in Relational DeepSym with an epistemic uncertainty reward computed from the covariance across a five-member forward-model council, addressing the bottleneck in the original paper: effect-prediction error grew with object count even as datasets scaled to 240K samples.
- Raised planning accuracy from **52% to 62%** over the random-exploration baseline at a matched budget of 51k samples, and reached 58% on 39k — 23% fewer than the baseline needed for 52%. The policy also reached interactions absent from the action repertoire, including rotation of a block through a controlled collapse.
- An earlier formulation searching for novelty in symbolic rather than effect space gained nothing over random exploration, because the encoder maps genuinely novel states onto familiar symbols; reformulating from *which states are novel* to *which are predicted poorly* removed that dependence.

Research Intern — SISReC, Osaka University, Japan — June – Sept 2023

Symbiotic Intelligent Systems Research Center · Funded by Osaka University's International Joint Research Promotion Program · Supervised by Prof. Erhan Öztop

Developmental Scaffolding with Large Language Models (*ICDL 2023*) — *publication page*

- Realized a study proposed by Prof. Erhan Öztop and carried it to publication as first and corresponding author. Designed a zero-shot, token-efficient state serialization converting scene configurations into LLM prompts, co-developed the PyBullet UR10 simulation, and ran the experiments against a random baseline, which surfaced systematic failures in GPT-3.5's affordance reasoning. Drafted the manuscript and handled the ICDL review response.

Human-to-Robot Skill Transfer through Correspondence Learning — project page

- On a suggestion from Prof. Minoru Asada, extended the CNMP-based correspondence-learning framework of Aktaş et al. (RA-L 2024, then in progress) from robot-to-robot transfer to a camera-tracked human demonstrator. Diagnosed the encoder's sensitivity to the low-frequency drift vision tracking introduces, distinct from the jitter the architecture tolerates, and corrected it with a filtering and normalization pipeline; cross-embodiment transfer to a simulated Torobo manipulator reached targets between the trained ones, largest observed error 2.98 cm.

Undergraduate Research Intern — CoLoRs Lab, Boğaziçi University — Aug 2022 – May 2023

Six-month TÜBİTAK STAR scholarship · Faculty supervision from Assoc. Prof. Emre Uğur; mentorship from Alper Ahmetoğlu, then a PhD candidate

Discovering Predictive Relational Object Symbols with Symbolic Attentive Layers (*RA-L 2024*) — *publication page*

- Designed and implemented the PyBullet simulation experiments, and showed that the preceding attentive architecture's failure to distinguish scenes holding two identical structures was structural rather than a matter of tuning. Built the automated experimentation pipeline, with Weights & Biases integration and VRAM-aware scheduling across 10–50 concurrent runs.

- Proposed and named the relational symbols formulation that addresses it, discretizing self-attention weights with Gumbel-Sigmoid into one sparse relational adjacency matrix per attention head. *The working implementation is my co-author Alper Ahmetoğlu's.*

TEACHING

Student Teaching Assistant — Boğaziçi University, Computer Engineering Feb 2022 – Jan 2025

- **CMPE443 Principles of Embedded Systems Design** (Fall 2024) — co-instructed the STM32 lab sessions and handled grading, for Prof. Arda Yurdakul.
- **CMPE250 Data Structures & Algorithms** (Spring 2023) — ran one term project end to end, from problem formulation through grading.
- **CMPE160 Object-Oriented Programming** (Fall 2023) and **CMPE150 Introduction to Computing** (Spring 2022) — problem sessions, question sets and exam problems.

COURSE PROJECTS

Both projects are joint work with Ilgaz Er, advised by Prof. Arda Yurdakul.

- **FPGA design space exploration via constraint-aware genetic algorithms** (CMPE583) — framed High-Level Synthesis parameter search as an integer-boolean optimization problem, mapping Vitis pragmas into chromosomes, with a selection rule that holds the population against a strict 8,000-LUT resource constraint. project page
- **Autonomous RC vehicle architecture** (CMPE443, 1st place in the course competition) — bare-metal C firmware for STM32 Nucleo-144 boards using direct register manipulation, without HAL dependencies.

TALKS & PRESENTATIONS

Developmental Scaffolding with Large Language Models — lab-exchange presentation, NAIST Robot Learning Lab, Aug 2023; poster, ROYAL Group, Boğaziçi, Nov 2023 · Reading-group presentations on NS-CL and PaLM-E, CoLoRs Lab, 2023

INDUSTRY & ENGINEERING EXPERIENCE

- **Freelance AI / Software Engineer**, remote (Jul 2024 – present) — LLM-backed conversational agents, a real-time video agent with speech synthesis and viseme-driven lip-sync, and a customer-support agent grounded in retrieved policy and account records.
- **AI / Software Engineer Contractor**, Upstash, remote / San Jose (Feb – May 2024) — developer advocacy for AI workloads: implemented GitFix and other demonstrations positioning Upstash's products as convenient infrastructure for AI applications; closed-beta tester for Upstash Vector.
- **Software Engineering Intern**, Atlassian, Ankara (June – Aug 2022) — OpsGenie's core distributed infrastructure on AWS primitives (S3, DynamoDB, SNS, SQS), and worked with teams in Poland and India to integrate the OpsGenie API with other Atlassian products.
- **IP Network & Automation Intern**, Nokia, Istanbul (Dec 2021 – Feb 2022) — internal tooling for remote ISP router configuration and infrastructure management.

TECHNICAL SKILLS

Research: experimental design · symbolic representation learning · structured world models · uncertainty estimation
Languages: Python · C++ · C and Embedded C · Java · TypeScript · JavaScript · SQL · Bash
ML & HPC: PyTorch · TensorFlow · CUDA · OpenMP · MPI · Weights & Biases · NumPy · PyBullet · ROS (familiar)
Systems & hardware: Docker · AWS · Redis · FastAPI · Git · Linux · FPGA (Xilinx) · STM32 · MATLAB/Simulink

HONORS & AWARDS

- **TÜBİTAK STAR Undergraduate Research Scholarship** — six months, supporting the work that became the RA-L 2024 study.
- **Ethics in Academia Delegate**, Bilkent University, 2016 — one of 150 students selected nationally, on assessed likelihood of pursuing a scientific career, to attend an academic-ethics lecture by Nobel laureate Aziz Sancar.

SERVICE

- **Departmental ABET accreditation representative**, Computer Engineering, Boğaziçi University — selected by two faculty to join the review interviews and coordinate senior-class participation.
- **Research computing**, CoLoRs Lab (Spring 2023) — ran the hardware evaluation for the lab's newly built GPU server on a roughly $12,000 budget, as the lab's parallel-computing and hardware hobbyist: compared candidate CPUs, GPUs and motherboards on architecture and fit to the lab's workloads. Presented the comparison to the lab, and helped bring the machine up on Linux, working mainly on the GPU and Ethernet drivers.
- **Weekly research seminar organizer**, CoLoRs Lab (Spring 2023); mentored incoming students in deep learning and experimental design.

LANGUAGES

Turkish (native) · English (full professional proficiency; TOEFL iBT 106 in 2023, retaking Dec 2026 ahead of the Fall 2027 cycle) · Japanese (conversational; classroom study at roughly N3)

REFERENCES

Dr. Alper Ahmetoğlu (CoLoRs Lab; co-author on both papers) · **Prof. Erhan Öztop** (Özyeğin / Osaka; SISReC supervisor, co-author on both) · **Prof. Suzan Üsküdarlı** (Boğaziçi; academic mentor) · **Prof. Arda Yurdakul** (Boğaziçi; both course projects). *Contact details on request.*